

# Aditya Pandey

Senior Software Engineer | Full Stack (.NET, React, Cloud, AI)

adityapandey241@gmail.com · Portfolio: [adiityaaaaaaaaa.github.io](https://adiityaaaaaaaaa.github.io)

## PROFESSIONAL SUMMARY

Senior Software Engineer with 2.5+ years of experience designing, building, and scaling distributed systems and cloud-native applications serving millions of concurrent users. Expert in .NET Core, ASP.NET Web API, React.js, TypeScript, microservices, CQRS, REST APIs, real-time communication, and event-driven systems. Proven track record of delivering business-critical systems under aggressive timelines, integrating AI into production, and optimizing performance at scale. Strong background in backend engineering, frontend development, system design, database optimization, cloud infrastructure, DevOps pipelines, and production reliability. Recognized with Employee of the Month for leading the design and delivery of a production-grade WhatsApp communication platform and AI chatbot. Actively seeking senior engineering roles in product-based companies, fintech, and global technology organizations.

## TECHNICAL SKILLS

- **Languages:** C, C++, C#, Python, TypeScript, SQL, Assembly (Beginner)
- **Backend:** .NET Core, ASP.NET Web API, Entity Framework, Microservices, CQRS, DDD, REST APIs, FastAPI, Redis
- **Frontend:** React.js, TypeScript, JavaScript (ES6+), Blazor Server, HTML5, CSS3, Responsive UI
- **Databases:** PostgreSQL, MongoDB, SQL Server, Redis, Query Optimization, Indexing, Data Modeling
- **Cloud:** AWS (EC2, S3, RDS, IAM, CloudWatch), Azure, GCP
- **DevOps:** Docker, Kubernetes (Familiar), Helm, CI/CD Pipelines, Octopus Deploy, Linux, Nginx, Git
- **Messaging & Realtime:** WebSockets, SignalR, Webhooks, Kafka (Familiar), RabbitMQ (Familiar)
- **AI & Automation:** OpenAI & Claude API Integration, AI Agents (tool use), RAG (Qdrant + embeddings), MCP Servers, Prompt Engineering, Chatbot Development
- **Architecture:** Distributed Systems, Event-Driven Architecture, Scalable Systems, API Design
- **Tools:** Postman, Studio 3T, Cursor AI, Claude Code, Jira, Confluence, GitHub
- **Soft Skills:** Strong Communication, High Ownership, Rapid Learner, Calm Under Pressure

## PROFESSIONAL EXPERIENCE

### Oakbrook Finance — Senior Software Engineer (Sep 2025 – Present)

- Led end-to-end design and delivery of a WhatsApp-based customer communication platform in under 2 months using .NET Core, React.js, TypeScript, WebSockets, and SignalR.
- Architected backend APIs, frontend UI, and real-time communication workflows for production scale and high reliability.
- Integrated a production-grade AI chatbot using OpenAI APIs, actively handling live customer queries in production.
- Drove the Open Banking VRP epic across three DDD/CQRS microservices — consent journeys, scheduled collection engine, HMAC-verified Modulr payment webhooks — plus the GoCardless direct-debit integration and bulk mandate import.
- Built a self-hosted CI/CD platform (.NET build engine + React/Vite SPA, Docker, Helm) that replaced hosted builds for the team, with SSE live logs, PR merge-preview builds and an MCP server for AI-assisted operations.
- Built AI support-desk tooling: GPT-powered ticket triage with auto-resolution, and a Claude-based agent using RAG over the codebase and ticket history with human-in-the-loop approvals.
- Designed authentication, authorization, logging, analytics, and monitoring systems to ensure security and observability.
- Optimized API performance, reduced response latency, and improved system throughput; implemented CI/CD pipelines and environment configurations for faster deployments.
- Recognized with **Employee of the Month** for rapid delivery and measurable business impact.

## Sportz Interactive — Backend Developer (Jul 2023 – Sep 2025)

- Built and maintained scalable backend systems using .NET Core, PostgreSQL, AWS, Redis, and monolithic architecture.
- Worked on large-scale sports platforms for global clients including F1, UEFA, and FEI.
- Designed REST APIs capable of handling 2–3 million concurrent users with low latency.
- Optimized database queries, indexing strategies, and caching layers to improve performance and reliability.
- Integrated AWS services including EC2, S3, RDS, and CloudWatch for monitoring and scalability.
- Handled production incidents, root cause analysis, and performance bottlenecks.
- Participated in system design discussions and contributed to architectural decisions.

## KEY PROJECTS & HIGHLIGHTS

---

Full case studies with UI previews: <https://adiityayaaaaaaa.github.io>

- **WhatsApp Customer Communication Platform** — Full-stack real-time messaging platform (.NET Core, React/TypeScript, SignalR, MongoDB, GCP) with a production OpenAI chatbot and live agent handoff; delivered in under 2 months.
- **SequensisCI — Self-Hosted CI/CD Platform** — 200+ commits: a self-contained .NET build engine (builds .NET/JS/R, publishes Docker/NuGet/Helm) under a React + Vite SPA with live logs over SSE; PR merge-preview builds, HMAC Bitbucket webhooks, atomic build versioning, Entra SSO, multi-replica scale-out with shared caches, and an MCP server so AI assistants can trigger and monitor builds.
- **CTSD-Agent — AI Support Agent** — Claude-powered tool-using agent that investigates and resolves support tickets using RAG (Qdrant + embeddings) over the codebase and historical tickets, with per-step human-in-the-loop approval; FastAPI + NiceGUI, Dockerised.
- **BatBot & BatBotUI — AI Ticket Triage** — Streamlit dashboard that AI-categorises Jira support tickets (GPT-4o-mini), enriches them from internal systems, and auto-processes whole ticket classes; Azure B2C OAuth2 PKCE; bulk back-office automation running 10 actions in parallel.
- **Open Banking VRP Platform** — ~300 commits across three .NET microservices: the six-screen consent journey with A/B rollout, a scheduled acquisition-collection engine, Modulr payment webhooks with HMAC verification, temporary-arrangement/suspension lifecycle, and brand-specific contractual documents (DDD, CQRS, MongoDB).
- **GoCardless Direct Debit & Bulk Mandate Import** — HMAC-SHA256 webhook processing rebuilt as CQRS domain events over mandate/payment lifecycles (API as source of truth), plus a Hangfire bulk-import pipeline with per-funder tokens, CMS-based id resolution and DDD batch aggregates.
- **Icenet Bulk Actions Platform** — async job engine turning agreement spreadsheets into thousands of back-office operations: Mongo-queued jobs, drain workers with bounded parallelism, eight action executors, a live-progress React UI, completion emails, and the cross-service Istio policy fix that unblocked it.
- **UtilityService + Link Shortener** — founded a production .NET 8 microservice solo (URL shortener shipped end-to-end in a day), then re-architected it through four phases to platform standard: CQRS/MediatR, OpenTelemetry, typed SDK + Pub/Sub events, and Istio mesh auth.
- **Loan Journey MCP Server** — MCP server + FastAPI/OpenAPI layer exposing a 14-endpoint loan application journey to AI assistants; powers a "Loan Advisor" Custom GPT that takes a user from first question to a live offer.
- **LocalCI + Octopus Deployer** — Python build-and-deploy toolchain: local Docker builds pushed to GCR with auto-derived tags, Helm packaging, and a deploy CLI with live status tailing; later absorbed into SequensisCI.
- **Loan Application Journey Website** — Production React 17 + TypeScript SPA where borrowers apply for loans; built a fraud-reduction "name on account" prefill/confirm flow and the Open Banking VRP payment-setup screens within its backend-driven (HATEOAS) navigation.
- **High-Traffic Sports Platforms** — Contributed to platforms serving 2–3 million concurrent users for F1, UEFA and FEI.

## EDUCATION

---

**Bachelor of Engineering (B.E.)** — Atharva College of Engineering, Mumbai University (2019 – 2023) · CGPA: 9.0/10

## ADDITIONAL STRENGTHS

---

- Strong debugging and problem-solving; high ownership mindset with consistent early delivery; calm in production incidents
- Comfortable across frontend, backend, cloud and AI layers in high-traffic production environments; rapid adoption of new technologies
- Mentored junior engineers, contributed to code reviews; excellent communication and stakeholder management skills